Magpie Gripper V2 Assembly

Overview

The Magpie Gripper is a versatile, 3D-printed robotic end-effector designed for precision grasping and vision-based manipulation. Driven by Dynamixel AX-12A servo motors and controlled via an OpenRB-150 board, the gripper utilizes a four-bar linkage system (crank and rocker) to ensure the fingers remain parallel throughout their entire range of motion. Additionally, it integrates an Intel RealSense D405 depth camera, allowing for advanced computer vision and autonomous grasping applications.

Safety & General Guidelines

CAUTION: Disconnect all power sources before wiring or rewiring the AX-12A servos or the OpenRB-150 board. Connecting power with reversed polarity can permanently damage the servo electronics.

Observe the following precautions throughout assembly:

- Work on clean, static-safe, well-lit surface
- Handle the Intel RealSense D405 by its housing edges only – avoid touching lens or IR emitters
- Do not force fasteners. If a screw does not thread smoothly, back it out, verify alignment, and retry
- Support printed parts near the hole being fastened to avoid cracking thin PLA walls
- Keep fingers clear of jaw mechanism when the gripper is powered and under servo control
- The parallel linkage mechanism and the gripping fingers create severe pinch hazards. Never place your fingers or hands between the jaws or the mechanical linkages while the servos are powered.

Bill of Materials

Verify that every item below is present before starting assembly. Quantities are per single completed gripper.

ID	Item	Quantity
1	Top Base (PLA)	1
2	Bottom Base (PLA)	1
3	Servo Crank (PLA)	2
4	Servo Rocker (PLA)	2
5	Finger (PLA)	2
6	Camera Protector (PLA)	1
7	OpenRB-150 Board	1
8	Intel RealSense D405	1
9	AX-12A Servo Motor	2
10	3M 6mm standoff	4
11	3M 8mm standoff	4
12	Electrical Wire	2
13	3M Bolts	16
14	2M Nuts and Bolts	12
15	5M bearings	6
16	Grip Tape	1

Tools Required (Not Included)

- 2M and 3M Allen keys
- Small flat-head screw driver
- USB A to USB C cable for the OpenRB-150

Assembly Instructions

Step 1: Prepare the fingers

Begin by cutting the grip tape into two equal strips. Apply one strip firmly to the inner gripping surface of each finger. Ensure that there are no air bubbles and the edges are flush with the plastic.

Step 2: Install the AX-12A Servo motors

Align the two AX-12A servo motors into the designated mounting brackets on the bottom base. Ensure the servo splines (the rotating output shafts) are facing inwards. Secure each motor using 4 2M nuts and bolts respectively.

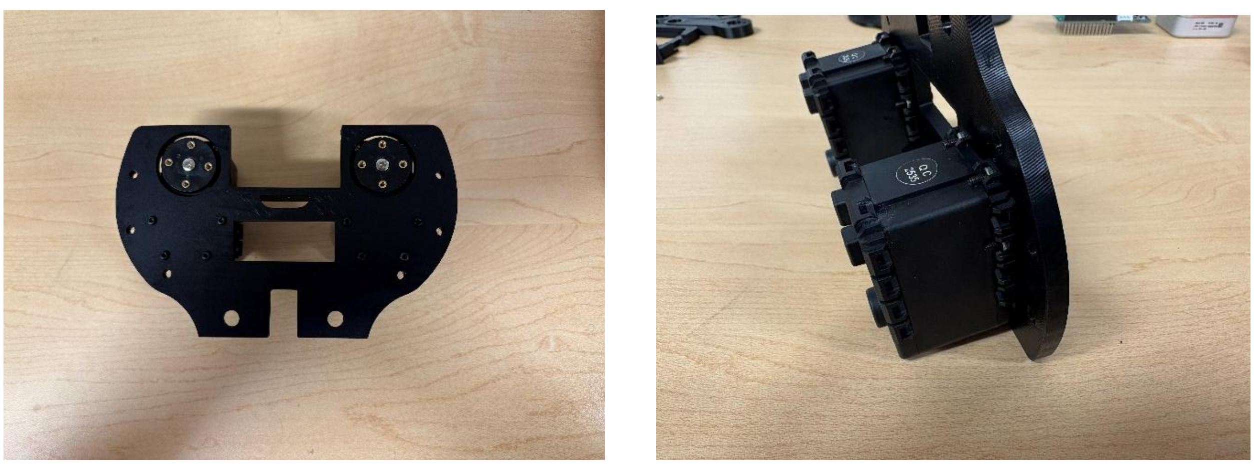


Figure 1 -- Left: Bottom base with both AX-12A servos installed, viewed from the bottom with both output horns visible. Right: the same pair of servos viewed edge-on, showing the flange screws that secure each to the housing.

Step 3: Build the crank & coupler linkage

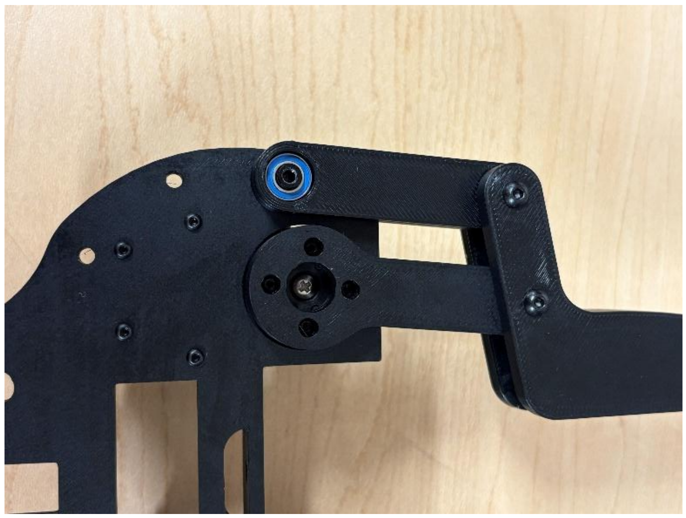
Press the 5M bearings into the circular housing slots on the servo cranks and servo rockers. Two bearings per rocker and one per servo crank

Align the mounting holes on the base of the fingers with the end-points of the servo cranks and rockers. Secure them using 3M bolts. Manually actuate the linkages to ensure smooth operation

Step 4: Mount the servo rockers and fingers

Attach a servo crank to the installed servo's output horn using the horn screw supplied with the AX-12A servo as well has 4 2M bolts. Make sure the notch of the servo is in line with the arm.

Connect the finger + coupler subassembly from step 3 to the crank's outer pivot with 3M bolts and a 8mm standoff, completing the four bar linkage on this side



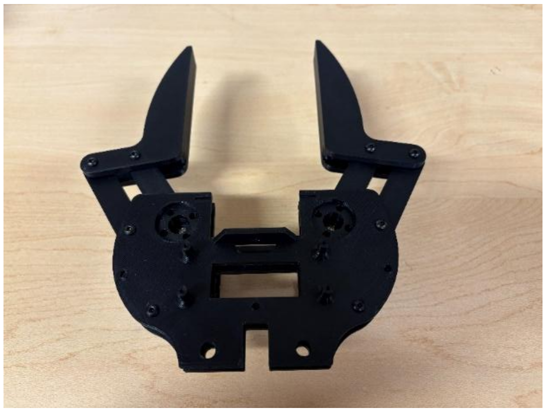
Step 5: Repeat the Linkage on the Second Side

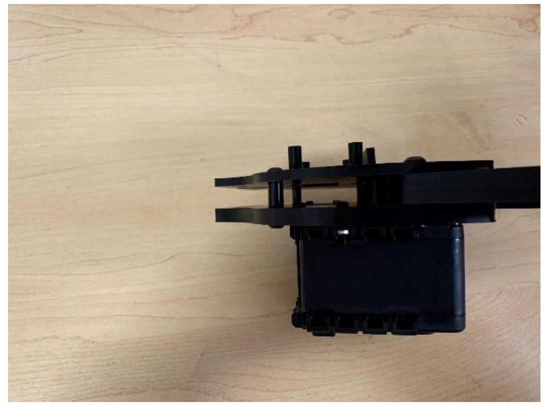
Repeat Steps 2 through 5 on the opposite side of the housing: lay out the second finger, crank, and coupler pair, pre-assemble the finger + coupler assembly, and attach it to the second servo's horn

With both sides complete, cycle each finger by hand to confirm smooth, symmetrical motion on both linkages before continuing.

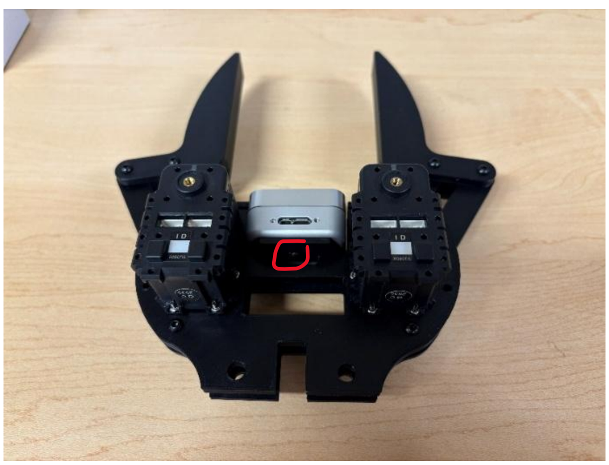
Step 6: Join the top base

Screw in 4 M3 screws to connect the top base to the rest of the assembly. There are additional holes for more standoffs and screws, but this is not necessary.





Step 7: Install the Intel RealSense D405 camera

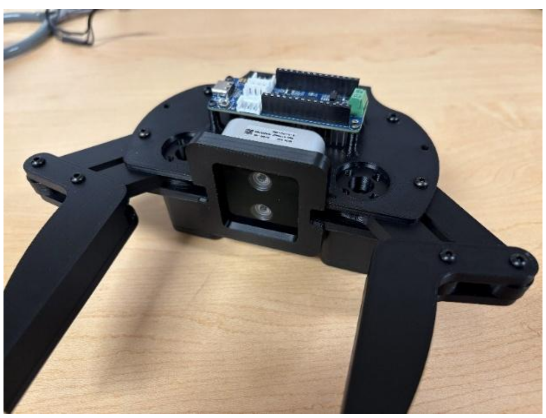


Make sure the camera orientation matches the figures with the cable facing the side of the servos. Do not use M3 screws larger than 6mm for this.

Step 8: Install the OpenRB-150 controller

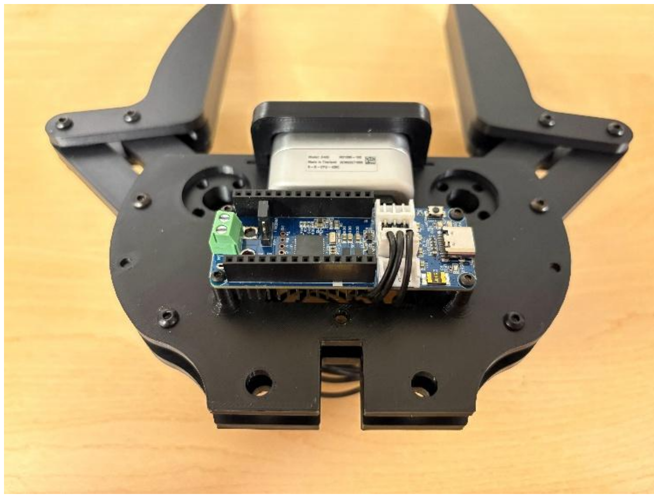
Use 4 M2 screws for this. Since the 3D printed part is not threaded, you may want to drive screws into the part before mounting the board.

Step 9: Install camera protector



Orientation of this part matters due to different sized top and bottom plates. Find the right orientation and snap in the part by applying reasonable pressure.

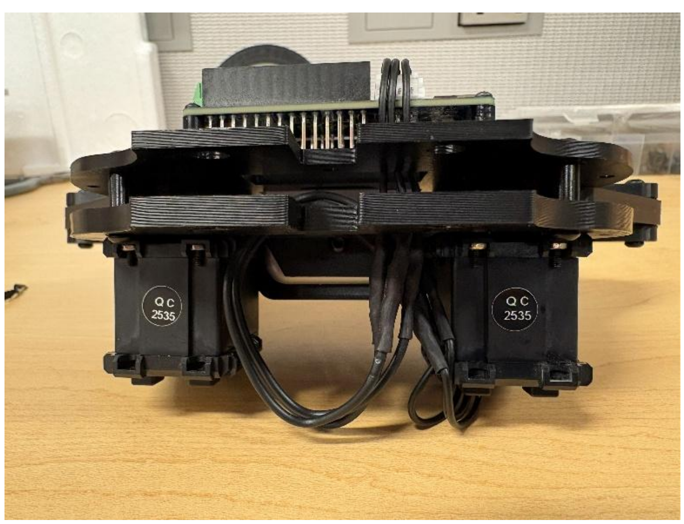
Step 10: Wiring and Cable Routing

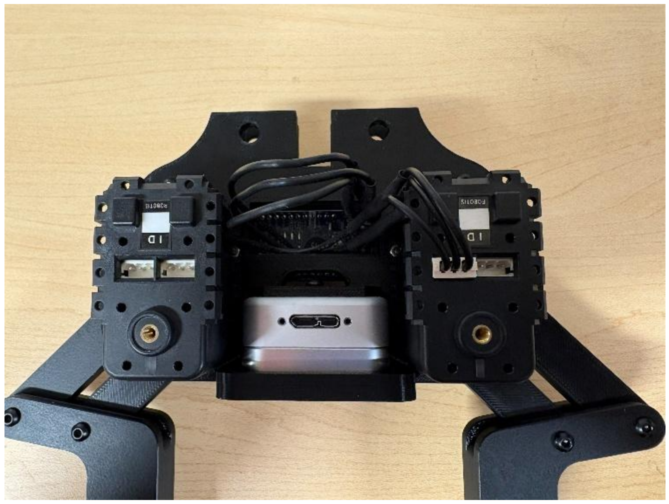


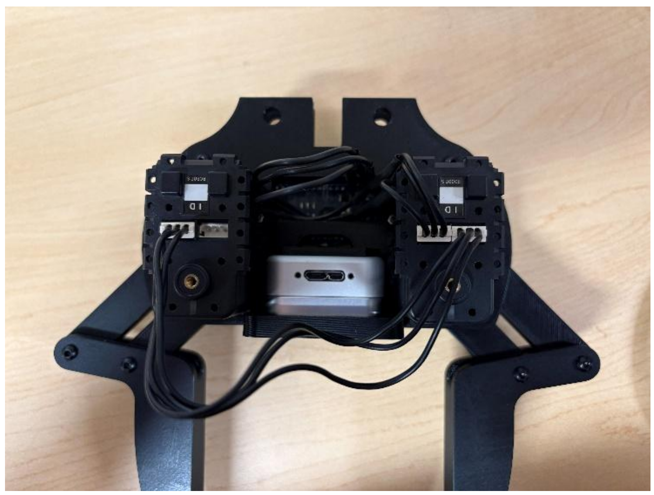
The EH to Molex cable should be routed through both base plates with the EH connector going to the OpenRB-150 board and the Molex connector to one of the servos.

Servos should be set with specific addresses with the right servo (in the orientation above with the board on top) set to ID 1, and the left servo set to ID 2. By default, servos are set to ID 1, so setting correct addresses may require connecting only one servo at a time. It is recommended to use Dynamixel Wizard 2.0 and to only connect the left servo first. While connected to a servo, max clockwise and counterclockwise positions can be set to minimize the chance of stalling a servo.

Also, the D405 camera should be calibrated by setting it up a known distance away from the surface and entering that information in. This can be done using the RealSense Viewer found in the RealSense repository.







Step 11: Final hardware check

Cycle each finger by hand (with gripper unpowered) through its full open and close range, confirming smooth motion with no binding at the servo coupler or rocker pivots

Recheck that all 3M and 2M fasteners used across Steps 1-7 are snug, and that no standoffs have backed out of the bottom base

Troubleshooting

Fingers do not move symmetrically

- Re-check that both servo cranks were installed with the servo horns centered before assembly. A horn installed off-center will cause one jaw to lead the other

Binding or rough motion in the linkage

- Loosen the affected servo coupler or rocker fastener slightly – pivots should rotate freely with zero to minimal axial play, not be clamped tight

OpenRB-150 does not detect a servo

- Verify the daisy-chain wire orientation and check for a firm connector seat at both the servo and the board. Confirm the servo's baud rate and ID match your control software's configuration

Camera not detected over USB

- Reseat the Intel RealSense D405 USB connector and confirm the cable screw was not overtightened, which can stress the internal cable

Calibration

For calibration of the RealSense D405 and the Dynamixel AX-12A servos, refer to step 10 of the assembly.

Maintenance

- Periodically inspect all 3M and 2M fasteners for loosening due to vibration and re-torque as needed
- Clean the Intel RealSense D405 lens with a microfiber cloth only – avoid solvents
- Check 5M bearings for smooth rotation every few hundred grip cycles; replace is roughness or play develops
- Store the gripper with fingers in the open position to minimize preload on the linkage when not in use